

Final Three-Seed Brightness Study

Real-only training versus a final mix containing 10% mild brightness-adjusted images

Design

Three paired random seeds: 42, 7, and 123. Each pair used identical model, image size, optimizer, epochs, data split, and training samples per epoch.

Question

Does adding 10% brightness-adjusted images improve segmentation performance under altered image brightness without harming clean validation performance?

Final conclusion

Do not treat 10% brightness augmentation as a confirmed robustness improvement.

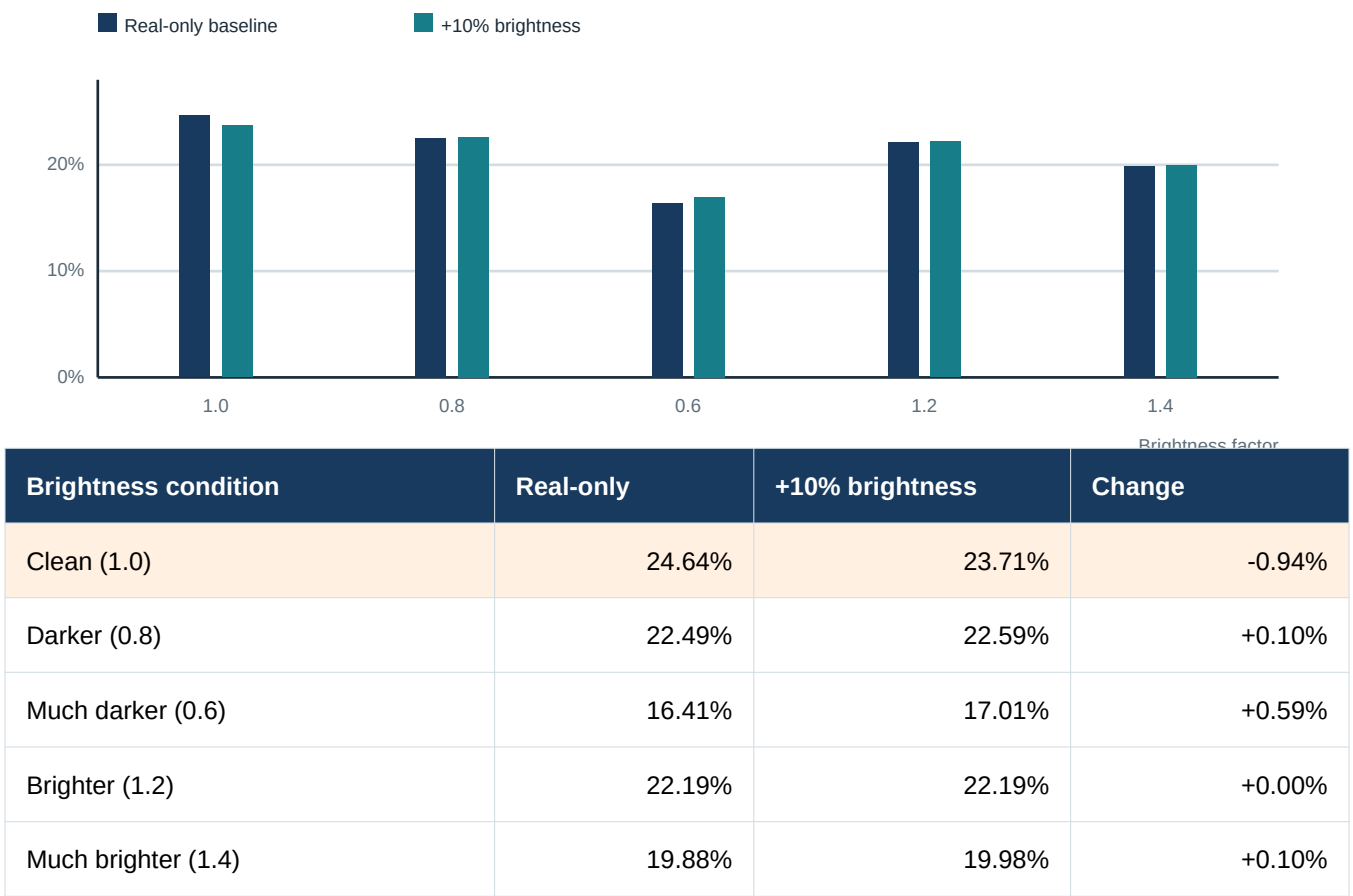
Across all three paired seeds, it consistently reduced clean validation mIoU by an average of 0.94 percentage points. Average robustness changes were close to zero and varied in direction by seed, so the apparent gains are not reliable enough to claim as an effect.

Fixed experiment settings

Setting	Value
Dataset	LoveDA Rural: 1,366 training pairs and 992 validation pairs
Model	Compact U-Net, 7 trainable classes, base channels 16
Image size	512 x 512
Training	15 epochs, Adam, learning rate 1e-4, batch size 1
Synthetic setup	10% final mix; brightness factors 0.8 and 1.2; labels unchanged
Evaluation	Frozen best checkpoints under five brightness factors, mIoU primary metric

Three-Seed Average Results

Mean validation mIoU. Higher is better. Changes are percentage points relative to the matched real-only model.



Why the final conclusion changed

The first two seeds suggested a possible benefit for severe darkening. The third seed reversed that pattern: its brightness model performed worse under both darker conditions. When all three seeds are averaged, the robustness changes are very small: +0.10 points at brightness 0.8, +0.59 at 0.6, 0.00 at 1.2, and +0.10 at 1.4. The clean-image decrease is more trustworthy because it happened in every paired seed: -1.62, -0.64, and -0.54 percentage points. This is the main result of the completed experiment.

Seed Variability and Next Decision

A transparent record of why more brightness tuning is not the right next experiment.

Per-seed robustness changes

Condition	Seed 42	Seed 7	Seed 123	Three-seed mean
Clean (1.0)	-1.62%	-0.64%	-0.54%	-0.94%
Darker (0.8)	-0.67%	+3.05%	-2.07%	+0.10%
Much darker (0.6)	+3.32%	+3.26%	-4.80%	+0.59%
Brighter (1.2)	+0.71%	+0.07%	-0.77%	+0.00%
Much brighter (1.4)	+1.86%	+0.93%	-2.49%	+0.10%

What to retain

The experiment pipeline is validated: paired seeds, fixed training budgets, saved checkpoints, clean and shifted validation conditions, and reproducible JSON reports. The real-only model remains the reference baseline for future synthetic-data studies.

What to do next

Freeze this 10% brightness result as inconclusive for robustness and mildly harmful for clean validation. Do not test more brightness ratios. Design a different augmentation experiment with one isolated variable, such as contrast change or geometric rotation, and evaluate it with the same paired-seed protocol.

Source files

evaluation/brightness_robustness/results.json; evaluation/brightness_robustness_seed7/results.json;
evaluation/brightness_robustness_seed123/results.json.